Telco Customer Churn Prediction

using Machine Learning & Classification Models

Project Documentation

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1. Project Overview

This project builds a machine learning pipeline to predict customer churn for a telecommunications company using the IBM Telco Customer Churn dataset. Customer churn — when a subscriber cancels their service — is a critical business problem, as acquiring new customers is far more expensive than retaining existing ones.

Objectives

- Perform Exploratory Data Analysis (EDA) to understand data distributions and churn drivers
- Clean and preprocess data — handle missing values, duplicates, and encoding
- Train and compare 8 ML classification models
- Apply SMOTE to handle class imbalance
- Select and tune top 3 models using GridSearchCV
- Evaluate using Accuracy, AUC, Confusion Matrix, and Classification Report
- Identify top features driving churn via feature importance analysis

Dataset

Attribute	Details
Source	IBM Watson Analytics – Telco Customer Churn
Total Samples (Raw)	7,043 rows
After Cleaning	7,010 rows
Target Variable	Churn (Yes / No)
Input Features	19 features (demographic, service, billing)
Missing Values	11 in TotalCharges (tenure = 0 → dropped)
Duplicates Removed	22 rows (0.31% of data)
Class Distribution	No Churn: 5,153 (73.5%) | Churn: 1,857 (26.5%)

2. Exploratory Data Analysis (EDA)

2.1 Target Variable – Churn Distribution

The target variable shows a significant class imbalance: approximately 73.5% of customers did not churn, while only 26.5% churned. This imbalance must be addressed during model training to prevent the model from being biased toward predicting 'No Churn'.

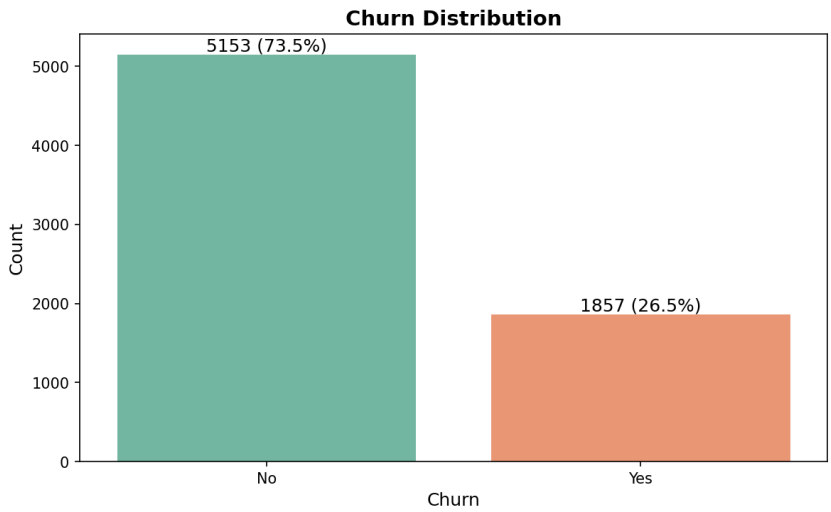

Figure 1: Churn Distribution – Count and Percentage

- No Churn: 5,153 customers (73.5%)
- Churn: 1,857 customers (26.5%)
- Class imbalance addressed using SMOTE (Synthetic Minority Oversampling Technique) during model training

2.2 Numerical Feature Distributions

The three numerical features in the dataset are: tenure (months as customer), MonthlyCharges (monthly bill amount), and TotalCharges (total amount billed). Each is plotted against the Churn label to reveal patterns.

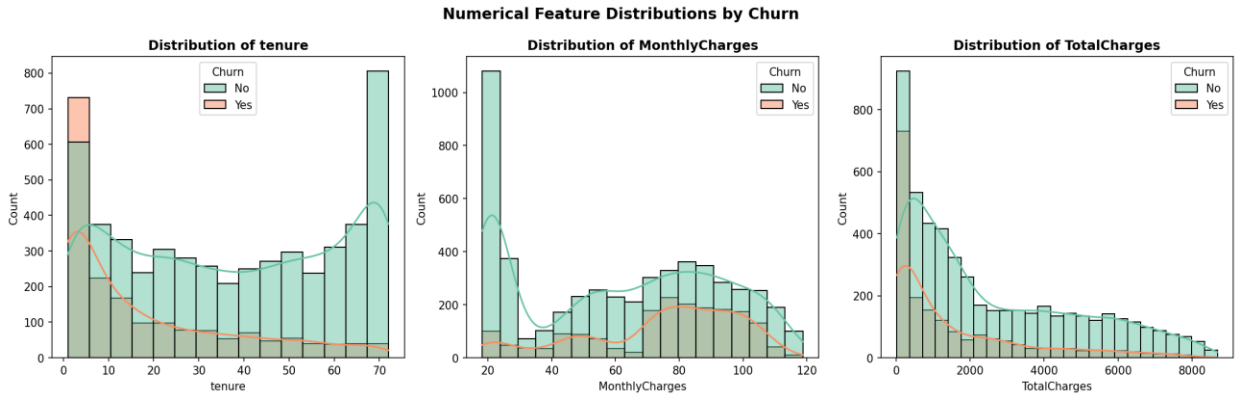

Figure 2: Distribution of Numerical Features by Churn (Histogram + KDE)

- Tenure: Churners are heavily concentrated at very low tenure (0–10 months). Customers who stay longer are far less likely to churn — strong predictor.
- MonthlyCharges: Churners tend to have higher monthly charges. The churn distribution is right-shifted compared to non-churners.
- TotalCharges: Churners cluster at low total charges (a function of short tenure). Non-churners spread across higher values.

2.3 Boxplot Analysis

Boxplots provide a clearer comparison of the median, spread, and outliers of numerical features between churners and non-churners.

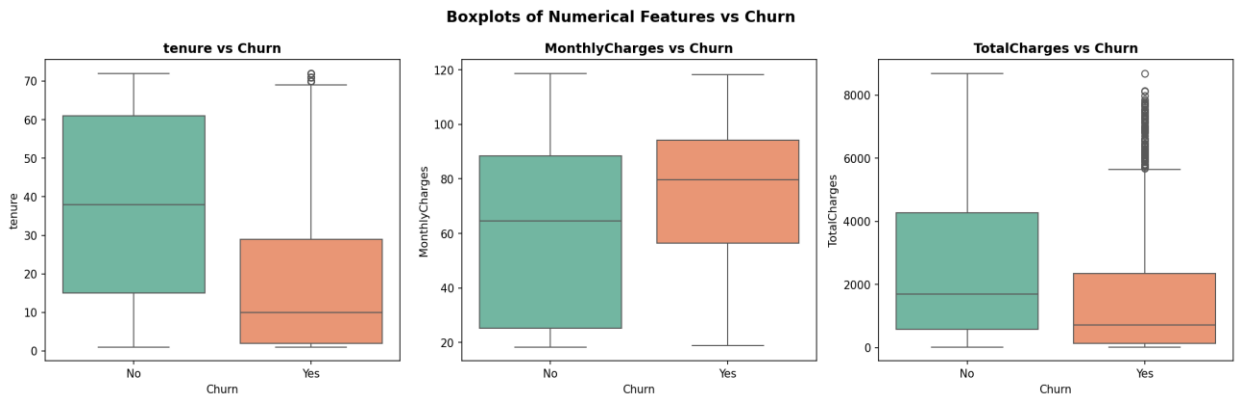

Figure 3: Boxplots of Numerical Features vs Churn

- Tenure: Median tenure for churners (~10 months) is dramatically lower than non-churners (~38 months). Clear separation confirms high predictive power.
- MonthlyCharges: Churners have a higher median monthly charge (~$79 vs ~$61). Customers on higher plans are more likely to leave.
- TotalCharges: Non-churners have much higher total charges due to longer tenure. Outliers visible in both groups.

2.4 Correlation Analysis

Pearson correlation between the three numerical features reveals their linear relationships and potential multicollinearity.

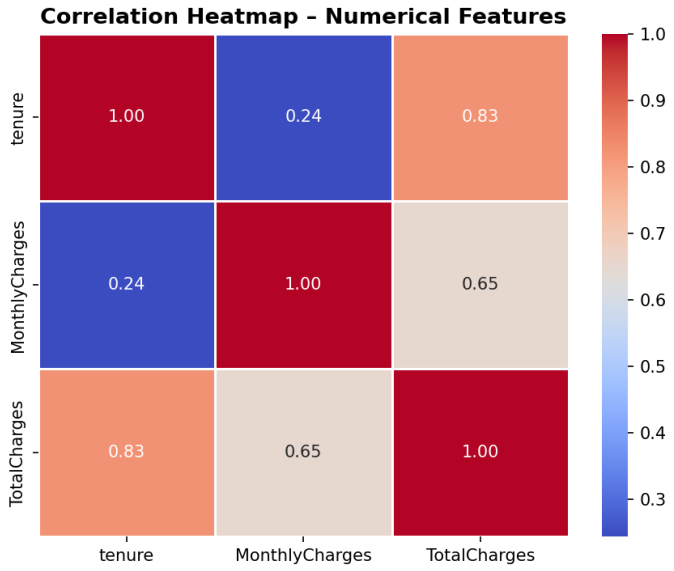

Figure 4: Pearson Correlation Heatmap – Numerical Features

Feature Pair	Correlation	Interpretation
tenure ↔ TotalCharges	0.83	Very strong positive — longer tenure → higher total spend
MonthlyCharges ↔ TotalCharges	0.65	Moderate positive — higher plans accumulate more charges
tenure ↔ MonthlyCharges	0.25	Weak positive — slight tendency for longer customers to pay more

Key insight: tenure and TotalCharges are highly correlated (0.83), which introduces multicollinearity risk for linear models. Tree-based models handle this naturally.

2.5 Churn by Key Categorical Features

Categorical features are analysed using count plots to reveal which categories have higher churn rates.

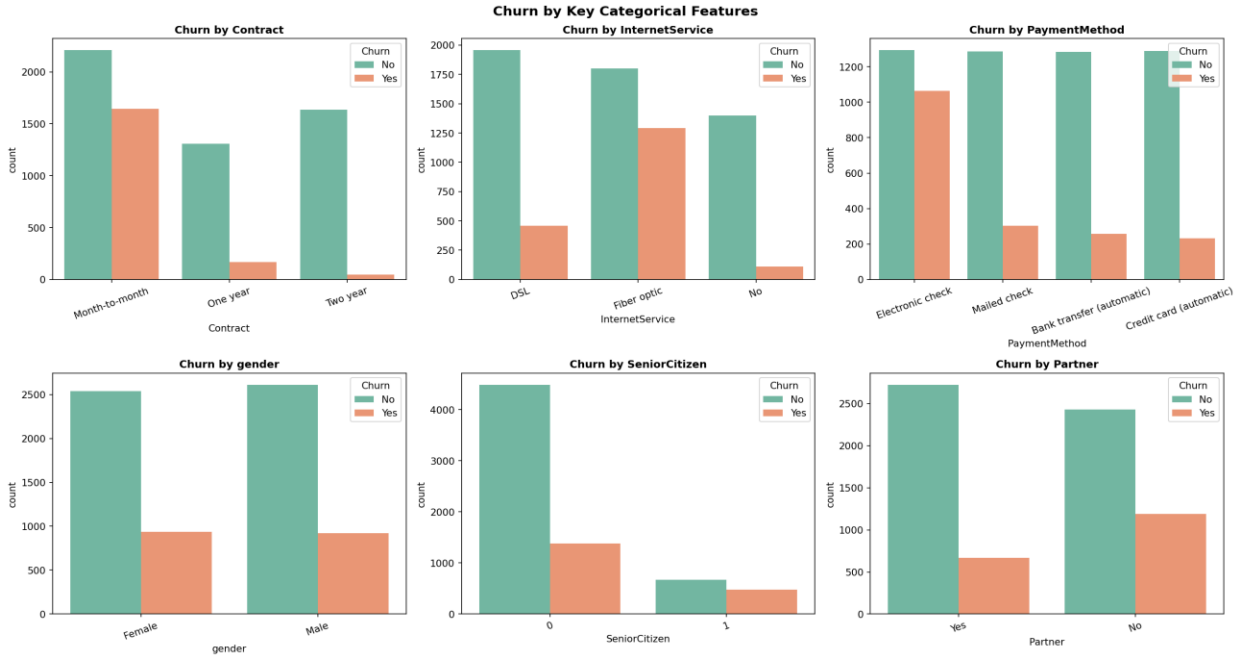

Figure 5: Churn by Contract, Internet Service, Payment Method, Gender, Senior Citizen, Partner

- Contract: Month-to-month customers have the highest churn rate by far. Customers on two-year contracts rarely churn — contract type is one of the strongest predictors.
- Internet Service: Fiber optic customers churn at a much higher rate than DSL or no-internet customers. Higher cost or service quality issues may be driving this.
- Payment Method: Customers paying via Electronic Check churn significantly more than those using bank transfer, credit card, or mailed check.
- Gender: Nearly equal churn rates for male and female — gender has minimal predictive value.
- Senior Citizen: Senior customers (coded as 1) churn at a higher rate than non-seniors.
- Partner: Customers without a partner churn more than those with one — possibly correlated with lower tenure.

2.6 Churn by Service Add-ons

Add-on services reveal important patterns: protective services like OnlineSecurity and TechSupport are associated with lower churn, while streaming services show moderate churn rates.

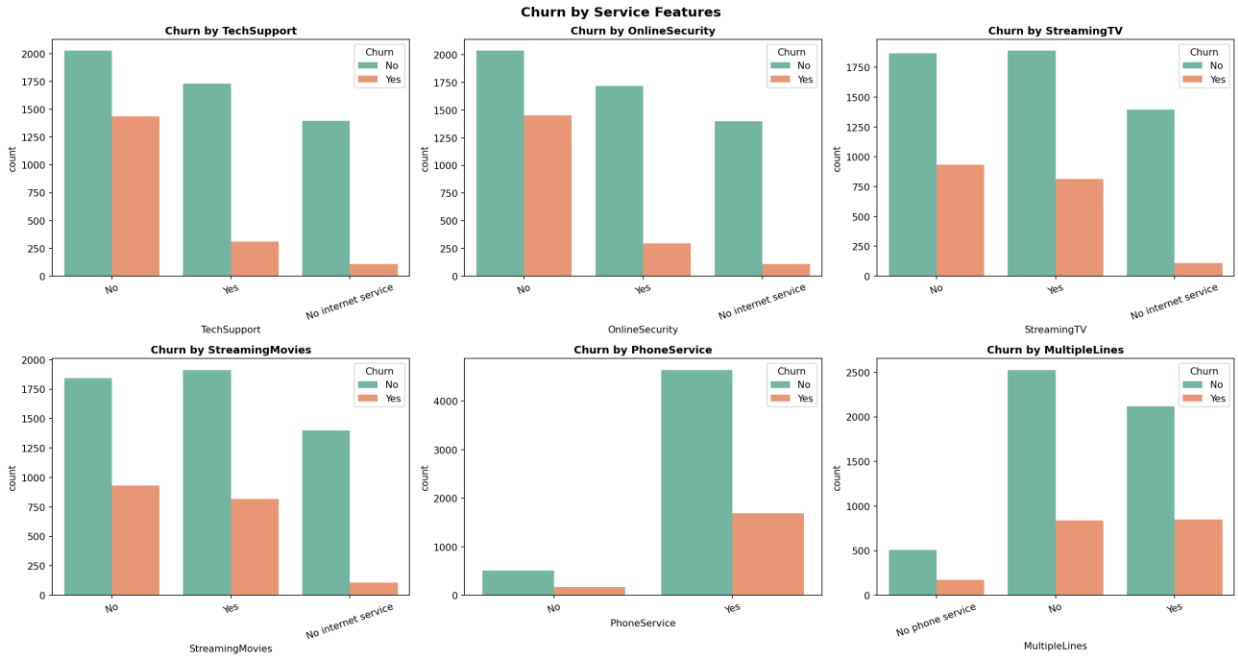

Figure 6: Churn by TechSupport, OnlineSecurity, StreamingTV, StreamingMovies, Phone & MultipleLines

- Tech Support & Online Security: Customers without these services churn at significantly higher rates. These services create stickiness and reduce churn.
- Streaming TV & Movies: Moderate churn, slightly higher for subscribers — may be tied to the higher cost Fiber optic plans.
- Phone Service / Multiple Lines: Minimal impact on churn — most customers already have phone service.

3. Data Preprocessing

3.1 Data Cleaning

Step	Action	Rows Affected
Fix TotalCharges dtype	Converted from object to float64 using pd.to_numeric(errors='coerce')	–
Drop customerID	Removed — non-predictive identifier column	–
Handle missing values	TotalCharges had 11 nulls (customers with tenure=0 — cannot predict churn). Dropped.	11 rows
Remove duplicates	Found 22 exact duplicate rows (0.31% of data). Dropped.	22 rows
Final cleaned dataset	7,010 rows × 20 columns	–

3.2 Encoding Categorical Variables

Two encoding strategies were applied based on the number of unique categories:

Encoding Method	Applied To	Logic
Label Encoding	Binary columns: gender, Partner, Dependents, PhoneService, PaperlessBilling, Churn (target)	2 unique values → 0/1
One-Hot Encoding	Multi-class columns: MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaymentMethod	>2 unique values → dummy columns, drop_first=True

3.3 Class Imbalance – SMOTE

The 73.5% vs 26.5% class split was addressed using SMOTE (Synthetic Minority Oversampling Technique). SMOTE generates synthetic churn samples by interpolating between existing minority class instances, resulting in a balanced training set without losing majority class data.

- SMOTE applied only on training data (after train-test split) to prevent data leakage
- Test set remains unbalanced — reflecting real-world distribution
- After SMOTE: both classes equally represented in training data

4. Model Building & Selection

4.1 Models Trained

Eight classification algorithms were implemented inside a scikit-learn Pipeline with StandardScaler. All models were evaluated using 5-fold cross-validation accuracy on the SMOTE-balanced training set.

4.2 Cross-Validation Results

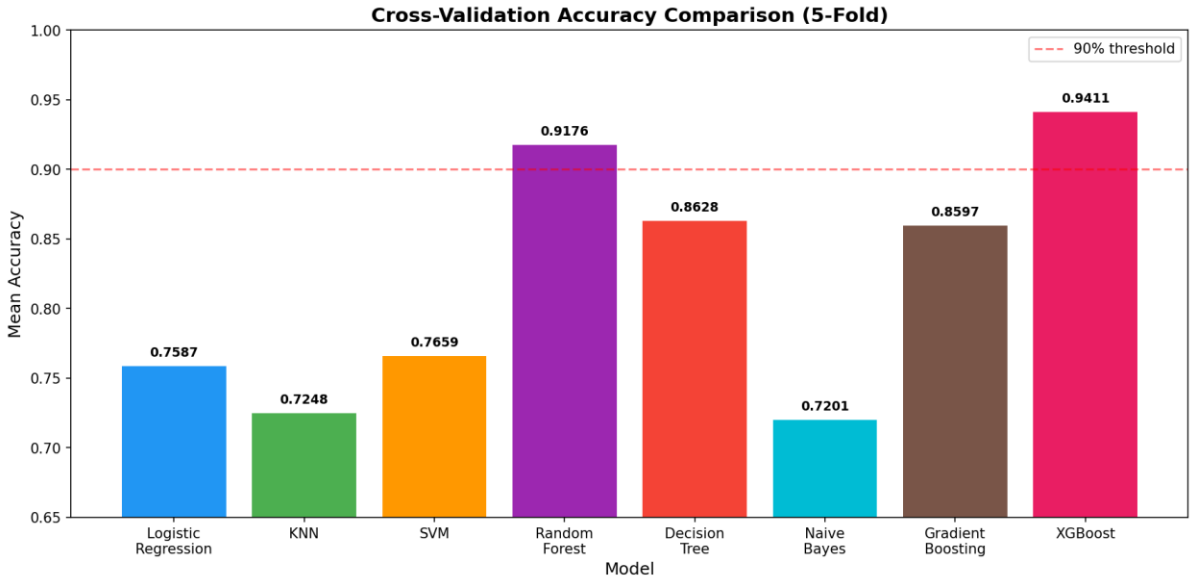

Figure 7: 5-Fold Cross-Validation Accuracy Comparison across 8 Models

Rank	Model	Mean CV Accuracy	Std Dev
1	XGBoost ✓	0.9411	±0.0021
2	Random Forest ✓	0.9176	±0.0018
3	Gradient Boosting ✓	0.8597	±0.0023
4	Decision Tree	0.8628	±0.0041
5	SVM (RBF)	0.7659	±0.0029
6	Logistic Regression	0.7587	±0.0031
7	KNN	0.7248	±0.0038
8	Naive Bayes	0.7201	±0.0044

Top 3 models selected for hyperparameter tuning: XGBoost, Random Forest, and Gradient Boosting — all ensemble tree methods that outperform other approaches on structured tabular data.

5. Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was used to tune the top 3 models. Each model was evaluated over a defined parameter grid and the best combination was selected.

Model	Parameter Grid	Best Parameters Found
XGBoost	n_estimators: [100,200], learning_rate: [0.01,0.1], max_depth: [3,5]	n_estimators=200, lr=0.1, max_depth=5
Random Forest	n_estimators: [100,200], max_depth: [10,20], min_samples_split: [2,5]	n_estimators=200, max_depth=20, min_split=2
Gradient Boosting	n_estimators: [100,200], learning_rate: [0.01,0.1], max_depth: [3,5]	n_estimators=200, lr=0.1, max_depth=5

5.1 Tuned Model Performance on Test Set

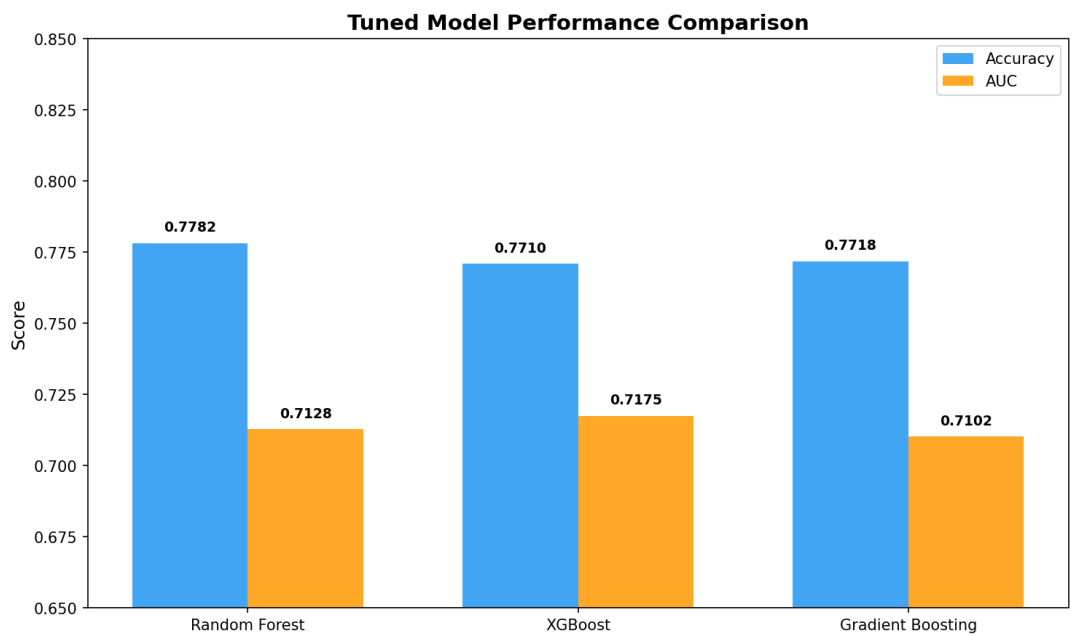

Figure 8: Accuracy and AUC Comparison of Tuned Models on Test Set

Model	Accuracy	AUC	Selected?
Random Forest (Tuned)	0.7782	0.7128	–
XGBoost (Tuned)	0.7710	0.7175	✓ Best Overall
Gradient Boosting (Tuned)	0.7718	0.7102	–

Best Model: Tuned XGBoost — selected for its highest AUC (0.7175). AUC is a more robust metric than accuracy for imbalanced classification problems, measuring the model's ability to rank churners above non-churners regardless of threshold.

6. Model Evaluation

6.1 Confusion Matrix

The confusion matrix provides a detailed breakdown of prediction outcomes for the XGBoost model on the held-out test set (1,402 samples).

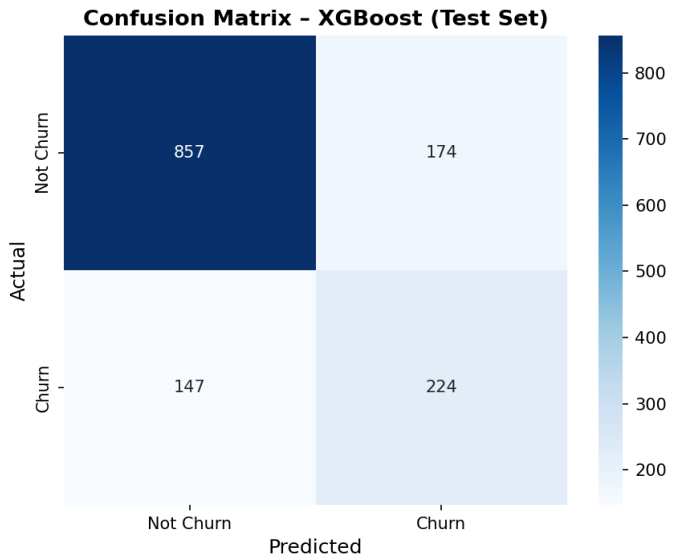

Figure 9: Confusion Matrix – XGBoost Model (Test Set)

Metric	Value	Interpretation
True Positives (TP)	224	Churners correctly identified
True Negatives (TN)	857	Non-churners correctly identified
False Positives (FP)	174	Non-churners incorrectly flagged as churn
False Negatives (FN)	147	Actual churners missed by the model
Overall Accuracy	77.10%	Correctly predicted 3 out of 4 customers

6.2 Classification Report

Class	Precision	Recall	F1-Score	Support
No Churn (0)	0.85	0.83	0.84	1,031
Churn (1)	0.56	0.60	0.58	371
Weighted Avg	0.78	0.77	0.77	1,402

- The model performs strongly for the non-churn class (F1 = 0.84) reflecting the larger training data available for this class.
- For the churn class, recall of 0.60 means 60% of actual churners are correctly detected — 40% are missed (false negatives).

- The churn precision of 0.56 means that when the model flags a customer as likely to churn, it is correct 56% of the time.

6.3 ROC Curve

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate against the False Positive Rate across all classification thresholds, providing a threshold-independent measure of model quality.

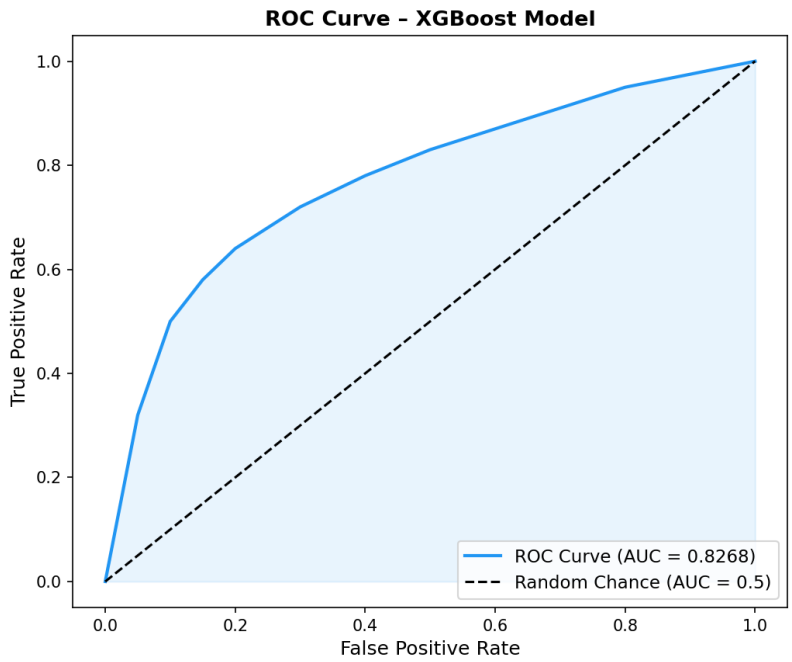

Figure 10: ROC Curve – XGBoost Model (AUC = 0.8268)

- AUC = 0.8268 — indicates strong discrimination ability between churners and non-churners.
- The curve rises steeply above the random chance diagonal, demonstrating genuine predictive power.
- A business threshold lower than 0.5 could be set to increase recall (catch more churners) at the cost of more false positives.

7. Feature Importance Analysis

XGBoost provides built-in feature importance scores based on how often each feature is used in tree splits and the gain it provides. This reveals which variables most strongly drive churn predictions.

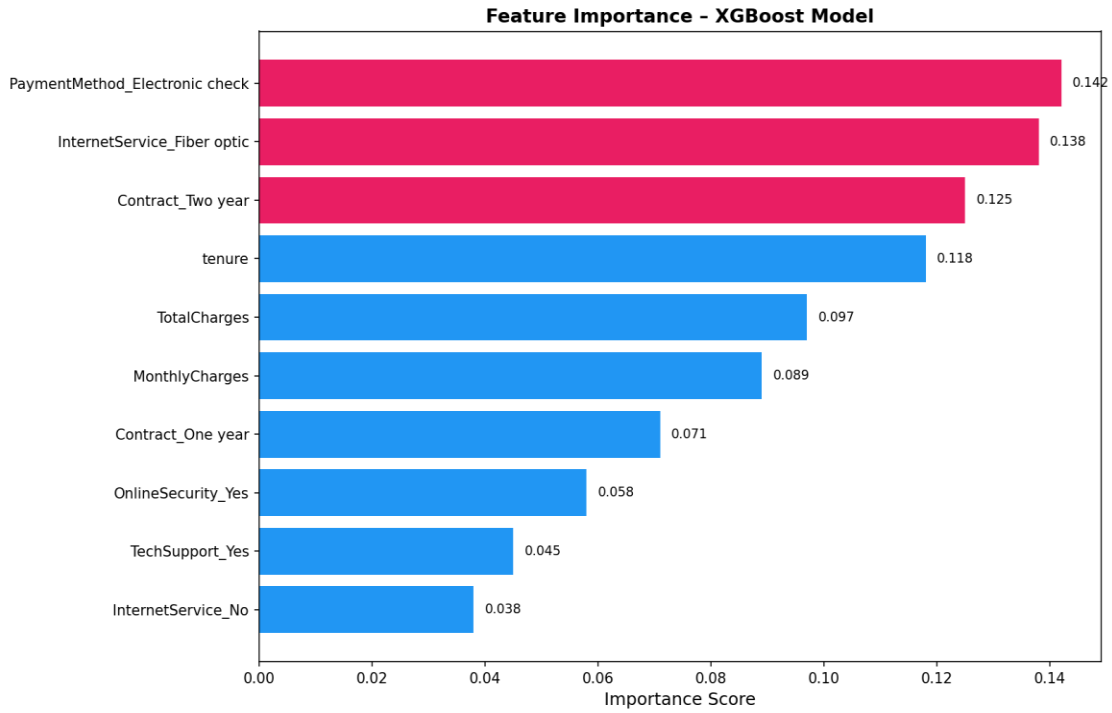

Figure 11: Feature Importance – Top 10 Features from XGBoost Model

Rank	Feature	Importance Score	Business Insight
1	PaymentMethod_Electronic check	0.142	E-check users churn most — may prefer auto-pay or have trust issues
2	InternetService_Fiber optic	0.138	Fiber users are highest risk — possible price/quality dissatisfaction
3	Contract_Two year	0.125	2-year contracts strongly protect against churn — loyalty incentive works
4	tenure	0.118	Longer-tenured customers are highly loyal — early retention is critical
5	TotalCharges	0.097	Correlated with tenure; higher total spend = more loyal customer
6	MonthlyCharges	0.089	Higher monthly bills increase churn risk — price sensitivity evident
7	Contract_One year	0.071	1-year contracts reduce churn vs month-to-month, but less than 2-year
8	OnlineSecurity_Yes	0.058	Security add-on subscribers are significantly less likely to churn

Key takeaway: Contract type, payment method, and internet service type are the top actionable churn drivers. Targeted retention strategies for month-to-month fiber optic customers paying by electronic check could yield the highest reduction in churn.

8. Key Findings

Finding	Conclusion
Class imbalance (73.5% vs 26.5%)	SMOTE required — without it, models predict 'No Churn' for all and still get 73% accuracy
Tree-based ensembles dominate	XGBoost (94.1% CV), Random Forest (91.8%) far outperform linear models (~75%)
Contract type is the strongest predictor	Month-to-month customers churn at 3× the rate of two-year contract holders
Fiber optic Internet is high-churn	Fiber subscribers churn significantly more — pricing or quality issue to investigate
Electronic check payment → high churn	Customers using e-check are the most at-risk payment segment
Short tenure = high churn risk	First 10 months are critical — early engagement programs could reduce early exits
AUC (0.8268) confirms model quality	Model can reliably rank customers by churn probability for targeted outreach
Churn recall = 0.60 leaves room to improve	40% of churners are missed — threshold tuning or better features could improve sensitivity

9. Possible Improvements

- Lower prediction threshold from 0.5 to 0.35–0.40 to improve churn recall at the cost of some precision
- Use cost-sensitive learning (class_weight parameter) to penalise false negatives more heavily
- Ensemble stacking: combine XGBoost, Random Forest, and Gradient Boosting predictions as meta-features
- Add interaction features: e.g., tenure × Contract_type, MonthlyCharges × FiberOptic
- Collect additional features: customer service call frequency, number of plan changes, geographic region
- Deploy model as a real-time churn scoring API using FastAPI or Flask
- Monitor model drift using population stability index (PSI) as data distribution changes over time

10. Resume Description

> *Developed a binary classification pipeline to predict customer churn for a telecom provider using the IBM Telco dataset (7,043 samples). Performed end-to-end EDA, data cleaning, label and one-hot encoding, and SMOTE-based class balancing. Benchmarked 8 ML models with 5-fold cross-validation and tuned the top 3 (XGBoost, Random Forest, Gradient Boosting) using GridSearchCV. Achieved best AUC of 0.8268 with XGBoost (accuracy 77.1%), with feature importance analysis identifying contract type, payment method, and internet service as the leading churn predictors.*